

PV239 – 04 IOC

Voting on bonus topics

- Closing - end of exercise **today**

Project status

- Teams
- Topic submission
- Discord
- Consultations

Exercise schedule

- Until 16. 3. - Submit questions for Microsoft
- 23. 3. **break** (no exercise)
- 30. 3. Last exercise + pub

Goals

- Get to know the principles of IoC/DI
- Passing arguments in Shell navigation

Goals

- Get to know the MVVM design pattern
- Look at the concepts for Data Binding

Dependency injection

- Filling dependencies “from the outside”
- Only declares what external dependencies it needs
- Dependencies are created outside of the class and have their own lifecycle
- Sample – class `IngredientListViewModel`
 - Needs to navigate to other pages – `IRoutingService`
 - Needs to load data - `IIngredientsClient`

Dependency injection

DEMO

IoC/DI

- Inversion of Control
 - Classes don't know about each other, communicate via interfaces
 - Classes don't create dependencies themselves
- Why? Boxes with exactly specified responsibilities
- They only take care of their business
- Can be replaced anytime with a different implementation
 - Even on the level of application configuration

IoC/DI in C#

- Class `ServiceCollection` + interface `IServiceCollection`
- Nuget package `Microsoft.Extensions.DependencyInjection`
- Container, where everything gets registered and from where it gets resolved

Lifetime management

- Singleton
 - 1 instance for the whole program lifetime
- Transient
 - New instance every time
- Scoped
 - The same instance for all classes within 1 request
 - A new instance for another request

Inversion of Control

DEMO

Shell – passing arguments

- Send parameters using the `Shell.Current.GoToAsync()` method
 - Can be part of Uri: `$"detail?id={id}"` - like HTTP request
 - Or passed as 2nd parameter:
`new Dictionary<string, object> { ["id"] = id }`
- On the receiving end
 - Add property to ViewModel
 - `public Guid Id { get; set; }`
 - Add class attribute to ViewModel
 - `[QueryProperty(nameof(Id), nameof(Id))]`

Shell – passing arguments

DEMO

Create a sample Ingredient edit page and add update functionality to IngredientsClient

DEMO

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